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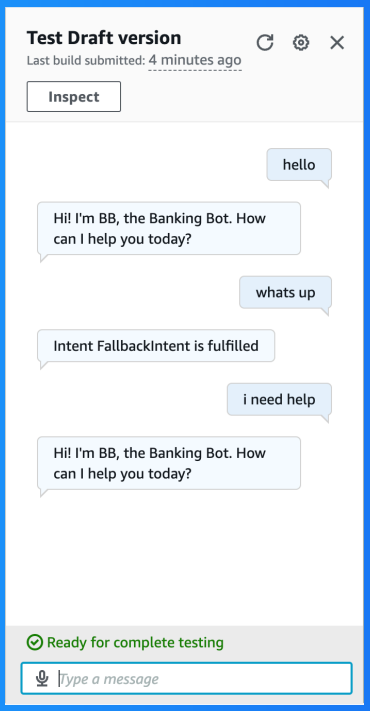
Build a Chatbot with Amazon Lex

James Paonessa



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Introducing Today's Project!

What is Amazon Lex?

Amazon Lex is a simplified chatbot system that is useful for a variety of applications that involve guiding users through an interaction to get to their results. If you understand the basic needs of the customer, this can speed up the process.

How I used Amazon Lex in this project

Today I used Amazon Lex to create a basic chatbot, BankerBot. It had a basic set of greeting messages and simple set of Fallback (error) messages.

One thing I didn't expect in this project was...

I found the project interesting as it open up my eyes to how the basic chatbot I interact with regularly are built. Learning how these bots are structured and how you can increase their efficancy for the end user is an important factor.

This project took me...

This project took roughly 40 min. of time.

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Setting up a Lex chatbot

I created my chatbot from scratch with Amazon Lex. Setting it up took me a few minutes to go through the settings and explore the various options before deciding on these.

While creating my chatbot, I also created a role with basic permissions because the chatbot will need its own internal permissioned role to interact with out AWS instances and buckets.

n terms of the intent classification confidence score, I kept the default value of 0.40. This means the chatbot needs to feel 40% confident that it understands the users response. This feels like an appropriate starting point to not throw errors.

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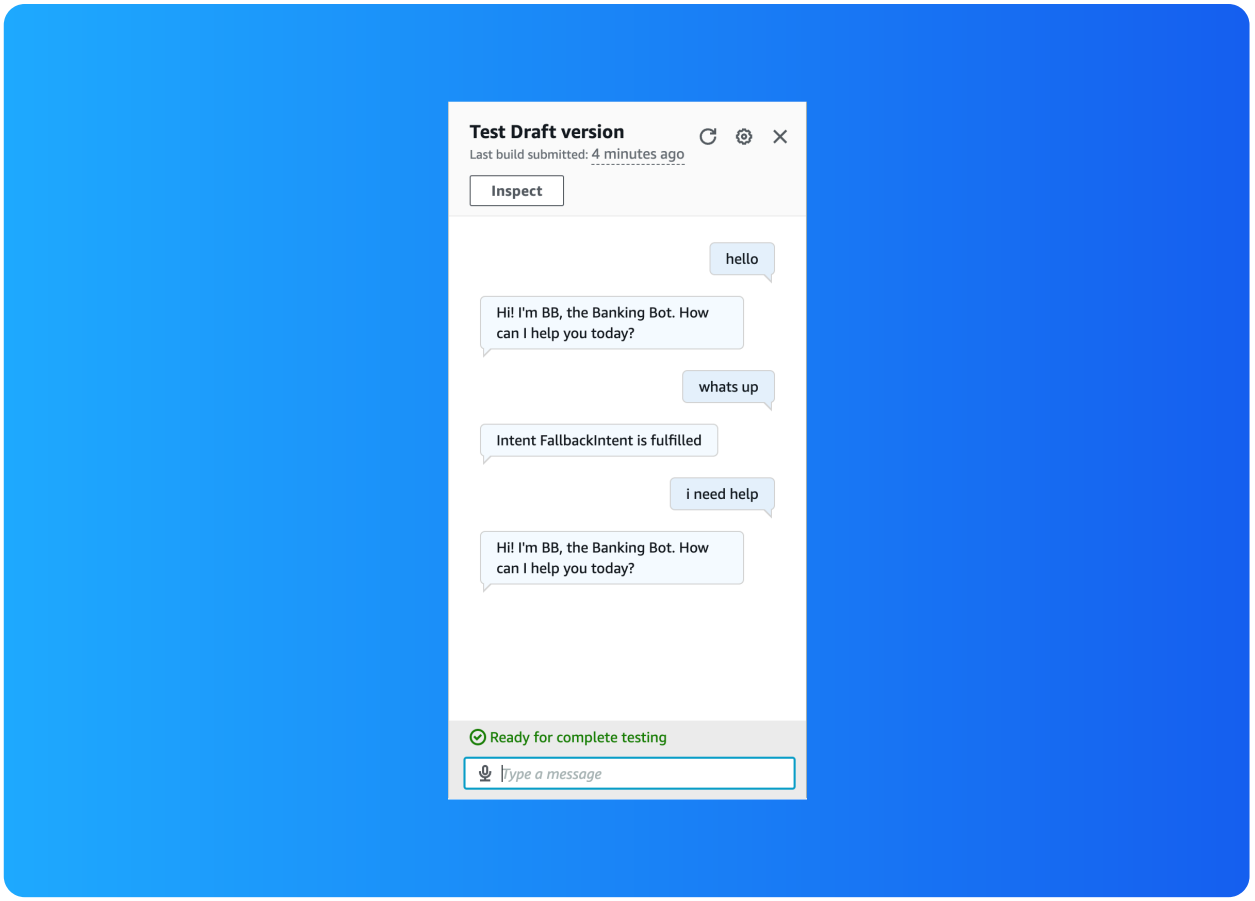
Intents

Intents are what the user is trying to achieve in the conversation. We can preprogram the chatbot to look for certain signals about users' intentions. For example; check balance, book flights, and check the weather.

I created my first intent, WelcomeIntent, to look for statements from the users that would indicate they are starting a conversation. This would prompt a message with the Bot's name and a welcome message.



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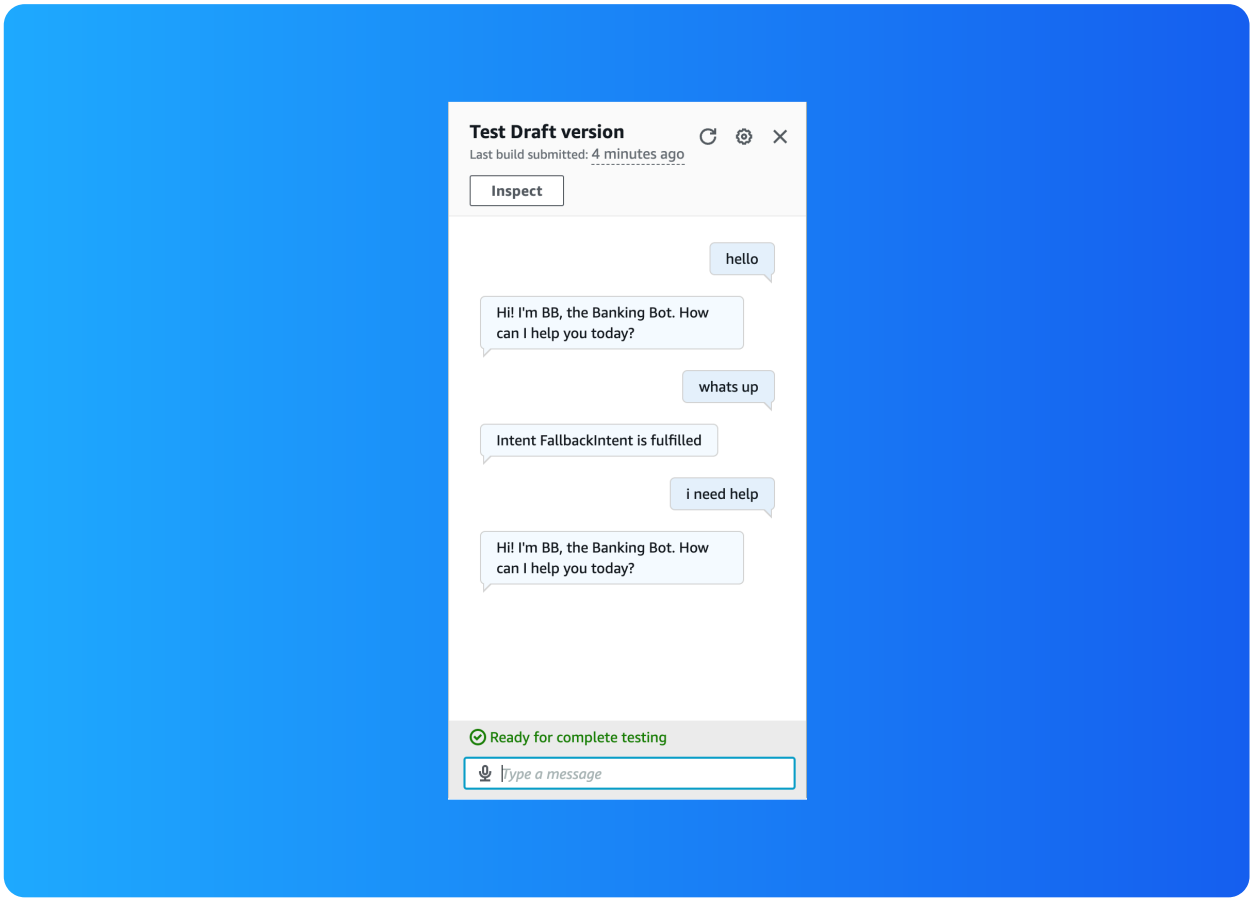
FallbackIntent

I launched and tested my chatbot, which could respond successfully if I entered inputs the chatbot was programmed to look for.

My chatbot returned the error message 'Intent FallbackIntent is fulfilled' when I entered "Whats up". This error message occurred because there was no preprogrammed response for this phrasing.



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Configuring FallbackIntent

FallbackIntent is a default intent in every chatbot that gets triggered when the chatbot's confidence level dips below the set threshold, which is set at 0.4 or 40%

I wanted to configure FallbackIntent to provide a more natural-sounding error message that could indicate to the users what options the might have to redirect them to a response that will work for them.

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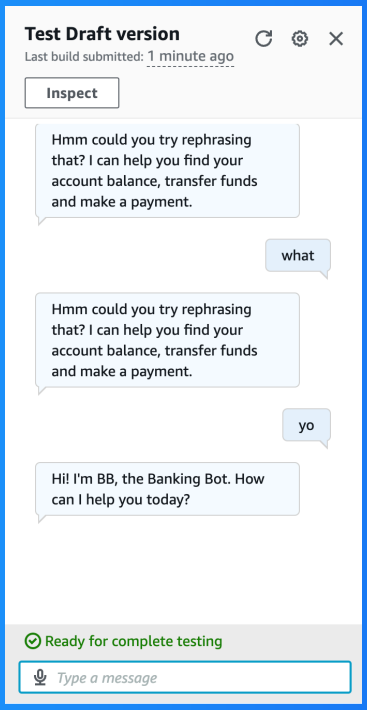
Variations

To configure FallbackIntent, I started by editing the closing responses of FallbackIntent.

I also added variations! What this means for an end user is there will be a subtle variation to the responses they will receive.